Logan Frank

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CONTACT

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EDUCATION

Ohio State University Columbus, Ohio
Ph.D. in Computer Science and Engineering
Advisor: Dr. Jim Davis
August 2019 - Present
Expected: Spring 2025

Wright State University, Dayton, Ohio
B.S. in Computer Engineering
Overall GPA: 3.88/4.00
August 2015 - May 2019

EXPERIENCE

Computer Vision Laboratory, Ohio State University, Columbus, Ohio
Graduate Research Associate
August 2019 - Present
Current: Long-Tail Image Classification
Previous: Data-Free Knowledge, Improvements to Batch Normalization, Computer Vision and Deep Learning in Agriculture

General Electric Research, Niskayuna, NY
Engineering Intern
May 2024 - August 2024
Mentor: Dr. Paul Ardis
Project: TBD!

ATR Center, Air Force Research Laboratory, Dayton, Ohio
Graduate Research Intern
May 2023 - August 2023
Mentor: Dr. Christopher Liberatore, RYAT
Project: Exploring Methods for Long-Tailed Image Classification

ATR Center, Air Force Research Laboratory, Dayton, Ohio
Graduate Research Intern
May 2019 - August 2019
Mentor: Mr. Christopher Menart, RYAT
Project: Utilizing Knowledge Graphs for Explainable Neural Networks

SMART Laboratory, Wright State University, Dayton, Ohio
Undergraduate Research Assistant
August 2018 - May 2019
PI: Dr. Yong Pei
Project: Small Object Detection System for Autonomous Aircraft

ATR Center, Air Force Research Laboratory, Dayton, Ohio
Undergraduate Research Intern
May 2018 - August 2018
Mentor: Ms. Ashley DeMange, RYDR
Project: Assessing Spiking Neural Network Performance on FPGAs

Marketo, Inc., Portland, Oregon
Quality Assurance Intern
June 2017 - August 2017

Research Interests

Computer Vision, Deep Learning, Generalizability / Robustness of Neural Networks, Adversarial Machine Learning, Self-Supervised Learning, Data Scarcity, Learning with Synthetic Imagery

Publications

L. Frank, J. Davis
"Data-Free Knowledge Distillation Using Adversarially Perturbed OpenGL Shader Images"
arXiv:2310.13782 [cs.CV], October 2023

J. Davis, **L. Frank**
"Revisiting Batch Norm Initialization"
European Conference on Computer Vision, 2022

L. Frank, C. Wiegman, J. Davis, S. Shearer
"Confidence-Driven Hierarchical Classification of Cultivated Plant Stresses"
IEEE/CVF Winter Conference on Applications of Computer Vision, 2021

Z. Daniels, **L. Frank**, C. Menart, M. Raymer, P. Hitzler
"A Framework for Explainable Deep Neural Models Using External Knowledge Graphs"
SPIE: Defense and Commercial Sensing, 2020

Awards

SMART Scholarship Recipient, U.S. Department of Defense
April 2019

Professional Service

Reviewer: CVPR'24, ICCV'23, CVPR'23, ECCV'22, CVPR'22
Mentored Students: Robert Sunderhaft, Christopher Cosma

Teaching

Software I: Components, The Ohio State University, Columbus, Ohio
Graduate Teaching Associate
Fall 2019, Spring 2020, Fall 2023, Spring 2024

"One of the best teachers I have had in my life, not just in college. Knew the content in and out, and taught with a true passion"
-Anonymous Student 1, Spring 2020

"Overall Logan Frank is one of the best professors I have encountered so far in my college career. He goes out of his way to make the content of the course better, providing additional resources of his own specifically for students, and is willing to help students at all times. He also just very friendly and a nice person."
-Anonymous Student 2, Spring 2020

"One of the best professors of my college career"
-Anonymous Student 3, Spring 2020

Circuit Analysis, Wright State University, Dayton, Ohio
Undergraduate Teaching Assistant
Spring 2019

Physics II: E & M, Wright State University, Dayton, Ohio
Undergraduate Teaching Assistant
Fall 2017, Spring 2018, Fall 2018

Other

NCAA Division 1 Athlete - Swimming, Wright State University
August 2015 - June 2017